

$$\|u - iv\|^2 - \|u + iv\|^2 = \cancel{\|u\|^2} + \cancel{\|v\|^2} + 2\operatorname{Im}\langle u, iv \rangle = \cancel{\|u\|^2} - \cancel{\|v\|^2} + 2\operatorname{Im}\langle u, iv \rangle$$

$$\|u - iv\|^2 - \|u + iv\|^2 = 4\operatorname{Im}\langle u, iv \rangle \quad | : 4$$

$$\frac{\|u - iv\|^2 - \|u + iv\|^2}{4} = \operatorname{Im}\langle u, iv \rangle$$

